3D Viewer Component – Technical Report

Module: Port Logistics – 3D Visualization | **Sprint:** B/C | **Date:** January 2026 | **Group:** 3DB-01

1. Assignment Description

The 3D Viewer is an interactive visualization module for the Port Logistics system, providing real-time 3D representation of port facilities and operational resources. It implements the following User Stories:

US 3.3.1 – Protected route access | **US 3.3.2** – Real data from Backend API | **US 3.3.3** – Display vessels and resources (STS, RTG cranes) | **US 3.3.4** – Appropriate textures/styling per element | **US 3.3.5** – Ambient + directional lighting | **US 3.3.6** – Mouse-based perspective camera | **US 4.2.1** – Select port facilities | **US 4.2.2** – Select operational resources | **US 4.2.3** – Role-based information overlay | **US 4.2.4** – Date/time simulation | **US 4.2.5** – Dynamic spotlight for selection | **US 4.2.6** – Smooth TWEEN.js animations | **US 4.2.7** – Camera reset

2. Technologies Used

Frontend: Angular 20.3.0, TypeScript 5.9.2, Three.js 0.181.1, TWEEN.js 25.0.0, RxJS 7.8.0
Backend: ASP.NET Core (REST API), Entity Framework Core
3D Assets: GLTF/GLB models, WebGL hardware-accelerated rendering

3. User Interaction

Mouse Controls

- **Left-Click:** Select port facility (dock, warehouse, yard) or operational resource (vessel, crane, truck)
- **Right-Click + Drag:** Orbit camera around scene or selected object
- **Mouse Wheel:** Zoom in/out

Keyboard Controls

- **W/A/S/D:** Move camera forward/left/backward/right (disabled during inspection mode)
- **I:** Toggle information overlay for selected object
- **R:** Reset camera to default overview position
- **ESC:** Deselect object and exit inspection mode

UI Controls

Layout selector (A/B), Reset View button, Night Mode toggle, Vessel Navigation animation, Animation Duration slider (200-3000ms), Easing Function selector, and Simulation Date/Time picker.

4. Architecture & Key Features

Component Structure: `viewer-3d.ts` (Angular component), `three.service.ts` (Three.js logic), `port-layout.service.ts` (API integration).

Camera System (US 3.3.6): Uses spherical coordinates (azimuth, elevation, distance) for smooth orbital control with min/max constraints.

Object Selection (US 4.2.1/4.2.2): Raycasting calculates normalized device coordinates, casts ray through click point, traverses hierarchy to find metadata, highlights with yellow outline and enters inspection mode. Selectable elements: Docks, Warehouses, Container Yards, Vessels, Cranes (STS, RTG), Trucks.

Role-Based Access (US 4.2.3): General data (name, type, dimensions) visible to all; restricted data (status, ETA, ETD, capacity) only for Port Authority Officers, Logistics Operators, and Administrators.

Animation System (US 4.2.6): TWEEN.js with configurable duration and easing functions (Linear, Quadratic, Cubic, Elastic, Bounce, etc.).

Lighting (US 3.3.5/4.2.5): Day/night mode with ambient + directional lighting, dynamic spotlight follows camera to illuminate selected elements (for groups with 4+ elements), visible penumbra effect.

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5. Visual Features & Additional Notes

Visual Features: Day/Night mode with sun/moon/port lights/lighthouse, fog effect for depth, reference grid and colored axes (X=Red, Y=Green, Z=Blue), yellow outline on selection, animated vessel arrival with audio feedback (ship horn), status colors for simulation (Purple=Arriving, Yellow=Waiting, Orange=Unloading, Blue=Loading, Green=Departing).

Technical Notes: Requires WebGL browser support, layouts dynamically loaded from backend API, model caching optimizes loading, proper resource disposal prevents memory leaks, protected route requires authentication.